

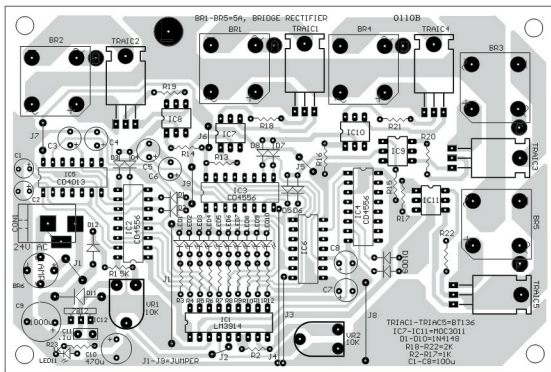


Fig. 5: Component layout for the PCB

traic and bridge rectifier. Here triac BT136 and 5A bridge rectifier are used for 1kW load. Triac BT139 with 10A bridge rectifier can be used for a solid-state relay of more than 1 kVA and less than 3 kVA. You can use up to 3 kVA solid-state voltage stabiliser with 3 kVA transformer.

Control power supply. Circuit diagram of control supply circuit is shown in Fig. 3. The 230V, 50Hz AC mains is stepped down by transformer X1 to deliver a secondary output of 24V, 500 mA. The transformer output is rectified by full-wave rectifier BR6, filtered by capacitor C9 and regulated by IC 7812 (IC12), which provides a 12V DC output. C10 and C11 provide further filtering. LED1 acts as the power indicator. Resistor R23 acts as a current limiter.

When capacitors used in the output are more than 10 μF , it is necessary to protect the regulator IC using diode (in this case, D11), in case their input is short to ground. Unregulated DC supply voltage is used for input sensing by IC1 for controlling the steps of mains transformer through solid-state relays.

Mains transformer. The mains

transformer used here is an auto-transformer with tapplings of 120V, 152V, 184V, 216V, 248V and 281V, respectively (as shown in Fig. 2). All the tapplings are connected with the voltage control solid-state relays to provide respective voltages. The tap at 216 volts is connected directly to the output.

Construction

An actual-size, single-side PCB for the solid-state voltage stabiliser is shown in Fig. 4 and its component layout in Fig. 5. The total circuit of solid-state voltage stabiliser can be assembled on a PCB. All the B7136 triacs need to be fixed on suitable heat-sinks with mica and insulated nut-bolts to isolate them. To begin with, the setting of solid-state voltage stabiliser is done without connecting mains transformer as described below.

1. Use a variable voltage transformer having 0 to 300 volts range and a digital voltmeter (3½-digit) for measurement of mains power supply.
2. Connect this solid-state voltage stabiliser with variable transformer starting from zero volts setting.

Increase voltage slowly from 0 to 120 volts

3. Set variable voltage transformer output of 120 volts with the help of digital voltmeter
4. Set low voltage setting pre-set VR2 of IC-LM3914 so that only LED1 glows
5. Now set the transformer at 281 volts with the help of digital voltmeter
6. Set high voltage setting pre-set VR1 of IC1 so that all LEDs from LED1 through LED10 can glow

7. Set the transformer at 184 volts and check the LEDs
 8. Set the transformer at 248 volts and check the LEDs
 9. Move variable transformer from 120 to upside and check solid-state relay output one by one with the help of a test lamp of 220V, 40W. Also see table of CD4556.
 10. Connect mains transformer tapplings to solid-state relays with great care at proper tapplings.
 11. Now connect digital voltmeter to output socket and check the voltage with variable transformer from 120V ~ 281V. The output should remain 220V
 12. Connect test lamp as a load and check voltage variation
- Now the solid-state voltage stabiliser is ready for use with a load of 1kVA.

EFY note. Male and female pin type power connectors may be used to connect mains transformer tapping with solid-state relays on PCB. Bar graph LED voltmeter connection also provides male and female pin connection from PCB to front panel of the stabiliser. ●